

Total No. of Questions : 10]

SEAT No. :

P3323

[Total No. of Pages : 3

[5353]-198

T.E. (Information Technology) (Semester - II)

OPERATING SYSTEM

(2012 Pattern)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume Suitable data if necessary.*

Q1) a) Explain micro kernel design approach? How will you decide that your requirement meets the criteria For micro kernel design? [5]

b) What resources are used when thread is created? How do they differ From those used when a process is created? [5]

OR

Q2) a) Explain the concept of Context Switching with the help of neat diagram. [5]

b) Discuss multilevel Feedback queue scheduling in UNIX. [5]

Q3) a) For the table given below, draw a Gantt chart illustrating the process execution using non preemptive priority scheduling. A larger no indicates higher priority. Calculate average waiting time. [5]

Process	Arrival Time	Burst Time	Priority
A	0	5	4
B	2	4	2
C	2	2	6
D	4	4	3

b) Explain message passing system For IPC and synchronization. [5]

OR

P.T.O.

- Q4) a)** Write the structure of producer and consumer process in bounded buffer problem using semaphore and discuss how critical section requirements are fulfilled. [5]
- b) Provide two programming examples in which multithreading provides better performance than a single-threaded solution. [5]

- Q5) a)** Consider the page reference string with 3 frames. [9]
A ,B, C, D, E, C, D, A, F, G, H, G, H, I, G, H, I, E, D, E, D, B
Calculate the no of page Faults For Following page replacement algorithms.
- i) FIFO
 - ii) OPTIMAL
 - iii) LRU
- b) Describe how Linux implements the following aspects of memory management. [9]
- i) Virtual memory addressing
 - ii) Page allocation
 - iii) Page replacement algorithm
 - iv) Kernel memory allocation

OR

- Q6) a)** Explain Belady's anomaly with suitable example. [4]
- b) What is cause of thrashing? How does the system detect thrashing? How the system can eliminate it? [6]
- c) Explain the address translation mechanism in paging and segmentation. [8]

- Q7) a)** Assume a disk with 200 tracks and the disk request queue has random requests in it as Follows: 98,183,37,122,14,124,65,67. Find the no of tracks traversed and average seek length If [8]
- i) FCFS
 - ii) SSTF
 - iii) SCAN
 - iv) C-SCAN is used and initially head is at track no 53.

- b) Explain different file organization techniques. [8]

OR

- Q8)** a) Why I/O buffering is needed? State and explain different approaches of I/O buffering. [6]
- b) Is disk scheduling, other than FCFS, useful in a single user environment. Explain u r answer. [6]
- c) What are different disk performance parameter? [4]
- Q9)** a) With neatly labelled diagram explain embedded Linux system architecture. [8]
- b) Explain following operations wrt NACH OS. [8]
- i) Modes of operations
- ii) Multiprogramming

OR

- Q10)** Write short notes on : [16]

- a) Ubuntu EDGE
- b) Android OS
- c) Service Oriented OS

